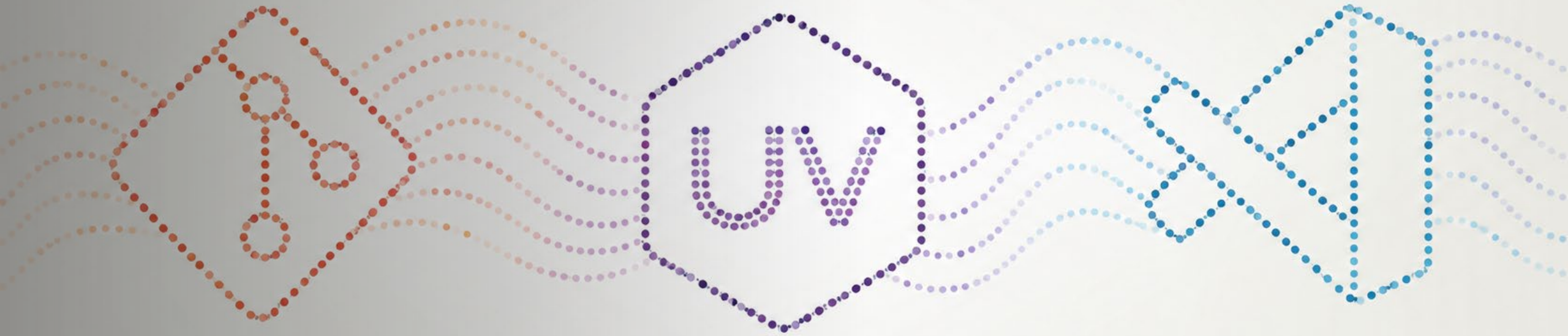

COURSE TOOLS



VS Code, Git, UV

VS CODE: INSTALL AND OPEN THE PROJECT

- Install from code.visualstudio.com/download

GIT: INSTALL AND CONFIGURE ONCE

- Install from git-scm.com/downloads
- Windows: Git for Windows • macOS: Xcode tools or installer • Linux: package manager

```
git --version
```

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your-email@example.edu"
```

GIT: THE EVERYDAY WORKFLOW

working files → stage → commit → GitHub

git add git commit git push

git status

git add path/to/file

git commit -m "Describe completed work"

git pull

git push

Use `git status` before every decision.

GIT: PROTECT A CLEAN HISTORY

- Commit small, meaningful milestones
 - Read `git status`; stage only intended files
 - Ignore local/generated files: `.venv/`, caches, raw data
 - If a push is rejected: save work → `git pull` → read carefully
 - If there is a conflict: pause and ask early if unsure
-

UV: INSTALL

Instructions: docs.astral.sh/uv

```
# Windows PowerShell
```

```
powershell -ExecutionPolicy ByPass -c "irm  
https://astral.sh/uv/install.ps1 | iex"
```

```
# macOS / Linux
```

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

```
uv --version
```

UV: LET THE PROJECT DEFINE THE ENVIRONMENT

```
pyproject.toml  →  dependencies and project settings
uv.lock         →  exact resolved versions
.venv/          →  local environment (do not commit)
uv sync         # match the repository environment
uv run ...      # run inside that environment
```

Avoid global `pip install` commands for a uv-managed course repository.

UV: THE NORMAL SESSION

```
uv sync
```

```
uv run python scripts/run.py
```

```
uv run pytest
```

```
uv run jupyter lab
```

Existing class repo:

```
git clone <repository-url>
```

```
cd <repository-folder>
```

```
uv sync
```

~~New project: git init → uv init → uv sync~~

REPRODUCIBILITY

Open repository root in VS Code

↓

`uv sync` → work → `uv run scripts/tests`

↓

`git status` → stage → `commit` → push

- `src/` for reusable code • `scripts/` for runnable workflows • `tests/` for checks
 - Keep `README.md` current: goal, setup, run command, file map
-

BEFORE SUBMITTING LABS/PROJECTS

1. Opened the repository root
2. `uv sync` completed; interpreter/kernel is correct
3. Work runs with `uv run ...`
4. `git status` shows only intended files
5. Meaningful commit created and `git push` succeeded

When stuck, share your OS, exact command, full error, tool versions, and `git status`.
